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National University of Computer and Emerging Sciences, Lahore Campus

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| Course: | Applied Thermodynamics | Course Code: | ME2002 |
| Program: | BS(Electrical Engineering) | Semester: | F 2024 |
| Duration: | 60 Minutes | Total Marks: | 40 |
| Paper Date: | 21-Sep-24 | Weight | 15% |
| Section: | ALL | Page(s): | 4 |
| Exam: | Mid1 Examination | | |

Name: [redacted]

**Instruction/Notes:**

- Use valid assumptions if needed, don't ask any questions.
- Your answers should be correct up to three decimal places with appropriate units.
- Values read from tables should have the same number of significant figures as mentioned in the tables.
- This is an open book, and open notes exam

## CLO 1: Analyze a thermodynamics process

**Q1** A closed system contains 5 kg of water in a Boiler. Initially, the water has a temperature of 25°C and a specific volume of $1.25616\ m^3/kg$. The system operates in two stages: **[20]**

- **Stage 1:** The system is heated until the pressure inside the boiler reaches 200 kPa keeping volume constant.
- **Stage 2:** The system is then cooled from a temperature of 150°C, allowing the pressure and temperature to change, but keeping volume constant.

a) Describe the phase and conditions of the water at $v = 1.25616\ m3/kg$ and $T = 25°C$?

b) When the pressure inside the boiler reaches 200 kPa, what will be the state of the water? Determine the corresponding temperature.

c) After the system cools from 150°C, what will be the total volume V of the water in the system? Describe the state of the water at this point.

d) If the water is a saturated mixture of vapor and liquid, what is its quality at $T = 100°C$? Also, illustrate this state and quality on a T−v diagram.

## CLO 1: Analyze a thermodynamics process

**Q2** Urea plants use ammonia as a working substance. In such a set up Ammonia is at $10°C$ with mass of 300 grams in a piston cylinder with an initial volume of $0.03\ m^3$. The piston has a design such that a pressure of 900 $kPa$ will float it. Now the ammonia is slowly heated to 50°C. Identify the states involved, find the final pressure and volume and draw this entire process on a Pv diagram. **[20]**

0.3 kg